ADIC LOCALIZATION AT AN ELEMENT PRIME TO P COMPUTES THE COMPLETION OF THE QUOTIENT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: Computes rig-closed points of adic localizations; the workhorse behind the rig-closed-point series in the rig geometry section.

Editorial note. The label lemma-rig-closed-point-in-localization names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Chapter notation: A with braces f denotes the restricted localization of the adic ring A at f, i.e. the completion of A_f with its adic topology
- Algebra, Section algebra-section-completion-noetherian: over a Noetherian ring the completion map is flat, the completion of a finite module M is M tensor the completed ring, and modules on which f is already a unit are complete

Source reference. Original section 0BNH: sources/stacks-chow/tags.txt, source lines 14665–14665. Prerequisite pointer invoked at 3483; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let A be an adic Noetherian topological ring. Let $\mathfrak{p} \subset A$ be a prime ideal. Let $f \in A$ be an element mapping to a unit in $A/\mathfrak{p}$. Then*

$$\mathfrak{p}A_{\{f\}} = \mathfrak{p}(A_f)^\wedge = \mathfrak{p} \otimes_A (A_f)^\wedge = (\mathfrak{p}_f)^\wedge$$

is a prime ideal with quotient

$$A/\mathfrak{p} = (A/\mathfrak{p}) \otimes_A (A_f)^\wedge = (A_f)^\wedge/\mathfrak{p}(A_f)^\wedge = A_{\{f\}}/\mathfrak{p}A_{\{f\}}$$

Proof. Since A_f is Noetherian the ring map $A \to A_f \to (A_f)^\wedge$ is flat. For any finite A-module M we see that $M \otimes_A (A_f)^\wedge$ is the completion of M_f. If f is a unit on M, then $M_f = M$ is already complete. See discussion in Algebra, Section 0BNH. From these observations the results follow easily. □

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